Maxwell Rosen

Undergraduate | UC San Diego

Southern California
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Education

University of California, San Diego, CA - *Bachelor of Science*

September 2024 - June 2028

- Chemical Engineering Major | Business Economics Minor
- **GPA: 3.96**
- Relevant Coursework: Thermodynamics, Material and Energy Balances, Organic Chemistry, Organizational Leadership, Project Management
- Marshall College Honors Program

Experience

Microfluidic Device Research - *Visiting Undergraduate Researcher*

National Taiwan University, Taipei, Taiwan • June 2026 - July 2026

- Selected for competitive UCEAP summer research at National Taiwan University to design, fabricate, and characterize microfluidic devices for biomedical applications
- Calculating saturation levels of a membrane by mass and imaging to optimize Western Blot experiments
- Developing an automatic micropipetting module for Western Blot and IHC

Salk Institute Research Intern - *Lab Intern*

Salk Institute, Ecker Lab • September 2025 - August 2026

- Designed and deployed 4 Raspberry Pi–based high-throughput imaging systems at 1% the cost of commercial alternatives ($500 vs. $50,000 per unit) to monitor plant growth across 200+ samples under variable environmental stress
- Collaborated on a 4-person team to troubleshoot and solve hardware-software integration challenges (image pipeline scaling, GPIO obstacles) while fulfilling time-sensitive project parameters
- Documented system architecture and authored setup protocols

Head Counselor & Inclusion Specialist - *MESSH Support, Inclusion/General Counselor*

Shalom Institute • Summers & Year Round 2023 - 2025

- Led a cabin of 10+ campers per cabin while delivering one-on-one support for the Mental, Emotional, Social, and Spiritual Health (MESSH) of 5 campers with special needs
- Designed and facilitated 20 programs per summer to promote values like building community, environmental stewardship, and role modeling

Skills

- Python, Raspberry Pi, Github, MATLAB, CAD Autodesk Fusion, Microsoft Excel